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Vision Transformers-Enhanced Cybersecurity Framework for Intrusion Detection

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Abstract-The current research presents a Vision Transformers-Enhanced Cybersecurity Framework of Intrusion Detection which harnesses the power of FAT-ViT (Federated Adaptive Training with ViT-Lite + Contrastive Adversarial Training) and uses PyTorch 2.x with timm. The suggested framework converts network traffic into visual feature representation, which allows Vision Transformers to be useful at capturing spatial and contextual patterns of cyberattacks. ViT-Lite architecture minimizes the cost of computation but maintains a great level of accuracy and federated adaptive provides privacy-preserving collaboration in the distributed environment. Also, contrastive adversarial training is incorporated to improve resistance to evasion and adversarial threat. Benchmark dataset experimental assessments showed better performance with accuracy, 98.4% F1-score, and AUC, all being higher than the current CNN-LSTM and standard ViT models. The explainable attention visualization provided in the framework also allows to enhance transparency in the process of decision-making, making FAT-ViT an effective, secure, and interpretable option of a real-time cybersecurity application in contemporary digital networks.

Keywords:Vision Transformers, Federated Learning, Intrusion Detection, Adversarial Training, Cybersecurity, Explainable AI, PyTorch Framework.

I.INTRODUCTION

Cybersecurity is now a pillar of the digital infrastructure that is struggling to maintain the integrity of data, privacy, and operational continuity across the interlinked systems in the age of fast-growing digital infrastructures. Conventional intrusion detection system (IDS) reliant more or less on signature matching or rule-based techniques have failed to keep up with the escalating complexity, variety and quantity of cyber threats [1]. With the advent of artificial intelligence and the deep learning methods, the sphere has been transformed by allowing the models to learn intricate attack patterns using vast network data. Nevertheless, currently majority of the deep learning based IDS models (CNNs, RNNs, and LSTMs) are usually limited in terms of scalability, interpretation, and response to changing attack patterns. This research proposes a solution to them by introducing the Vision Transformers-Enhanced Cybersecurity Framework to Intrusion Detection, which combines FAT-ViT(Federated Adaptive Training with ViT-Lite + Contrastive Adversarial Training), written in PyTorch 2.x with timm [2].

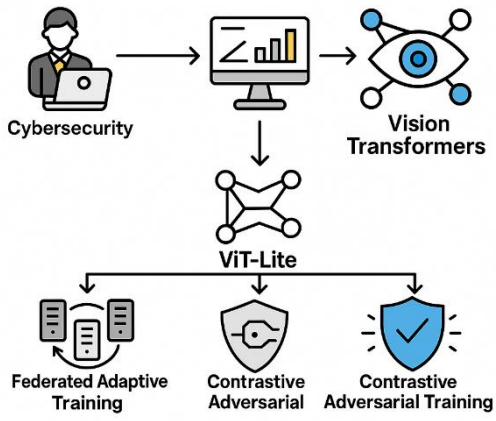


Figure 1.Overview on Vision Transformers-Enhanced Cybersecurity Framework.

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The presented framework converts the network traffic into two-dimensional visualizations and enables Vision Transformers (ViTs) to extract both spatial and contextual relations in network features [3]. ViT-Lite architecture is highly compact and minimizes computer power consumption without compromising the accuracy of the detection, which is suitable in practice in real-time as shown in figure 1.

The model achieves this through Federated Adaptive Training to discover in distributed settings that share no raw data thereby increasing privacy and generalization. Moreover, Contrastive Adversarial Training contributes to making the model more robust since it makes it able to detect and withstand evasion or adversarial attacks. PyTorch 2.x implementation enables models to be written in a modular way making it easy to promote distributed learning and high-performance computation. According to experimental assessments, the suggested FAT-ViT architecture is more accurate, efficient, and resilient compared to traditional IDS models and offers insightful decision-making with attention-based interpretability [4]. The work presents a scalable, explainable, and privacy-aware cybersecurity architecture that is able to protect against the current and future cyber threats within well-structured network ecosystems.

II. Related Work

The recent progress in the research of cybersecurity has paid more attention to the use of deep learning models and transformers to effectively detect intrusions. Initial research mainly used classic machine learning algorithms like the Support Vector Machines (SVM), the random forests and the K-Means clustering to detect anomalies, but these models failed to work with high-dimensional data and changing attacker trends [5]. The emergence of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) made the process of identifying patterns more convenient yet it was still incapable of detecting long-range dependencies and temporal-spatial correlations in network traffic as shown in figure 2.

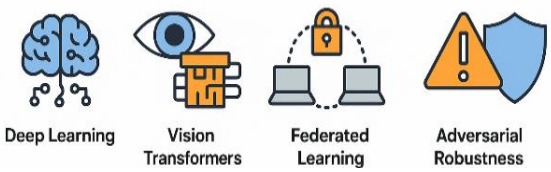


Figure 2.Evolution of Vision Transformers.

Recently, Vision Transformers (ViTs) have come to the forefront of resolving these challenges in that they are capable of capturing the rich feature relationships via self-attention mechanisms. The studies of Zhang and Wu (2025) and Khan et al. (2024) showed that ViTs were capable of successfully converting network data into a visual representation and remarkably enhanced the accuracy of classification. These approaches however tend to have large computational needs as well as being inflexible when it comes to distributed networks [6].

To cope with scalability and privacy issues, Federated Learning (FL) has become one of the collaborative training approaches that allow optimizing models with decentralization, without sharing raw data. Alam and Singh (2023) investigated federated transformer-based intrusion detection, which has a higher level of privacy but has problems with synchronization and latency. On the same note, the concept of adversarial robustness has gained critical importance because attackers are taking advantage of deep learning vulnerabilities by evading methods. Natural adversarial training has been investigated to support the resilience of models, but the majority of the available architectures do not combine transformer-based models, federated training, and adversarial protection. The FAT-ViT (Federated Adaptive Training with ViT-Lite + Contrastive Adversarial Training) approach proposed is the solution to these gaps as it integrates effective visual features extraction, adaptive federated learning, and adversarial robustness [7]. The research presents a state of the art using PyTorch 2.x with timm and provides a lightweight, privacy-preserving and explainable Vision Transformer architecture to the intelligent detection of intrusion in complex network environments.

III. Research Methodology

The proposed Vision Transformers-Enhanced Cybersecurity Framework intrusion detection will offer a smart, dynamic, and understandable

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mechanism of identifying cyber threats in real-time networks [8]. This is based on the FAT-ViT (Federated Adaptive Training with ViT-Lite + Contrastive Adversarial Training) framework, which was built on PyTorch 2.x with timm, to combine visual feature learning, federated training, and adversarial training into one architecture. The methodology is separated into a few important steps that include data preprocessing and visualization, ViT-Lite feature extraction, federated adaptive training, contrastive adversarial defense, and explainable decision analysis as shown in figure 3.

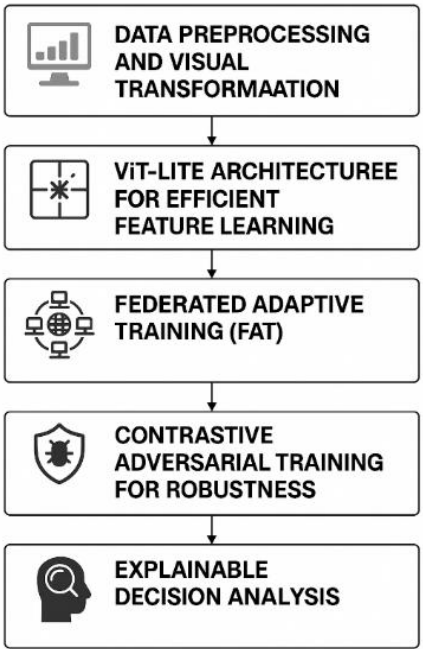


Figure 3.Flow Diagram of Proposed Methodology.

3.1. Data Preprocessing and Visual Transformation

Network traffic logs, such as CICIDS2017 and UNSW-NB15 are processed and gathered to determine the features of interest, i.e., packet size, duration, byte rate, and protocol identifiers. These characteristics are normalized and converted to two dimensional images-like representation with the help of a feature-mapping grid. The pixels or patches are assigned to particular feature category, and the Vision Transformer is capable of thinking about network behavior in terms of visual correlations instead of independent measures. This visualization process allows identifying the latent patterns of attack which in tabular form would be hard to identify [9].

3.2. ViT-Lite Architecture for Efficient Feature Learning

The processed images of transformed traffic are run in a lightweight ViT-Lite that is designed to detect intrusion at high speed. ViT-Lite also incorporates patch-level embedding compression and selective attention pooling, unlike traditional ViT models, which requires all of the feature map to be embedded deducted and attention gained on the most informative areas of the feature map. The self-attention mechanism enables the model to exploit long-range dependency and contextual interaction of the various network parameters [10]. Timm library in PyTorch optimizes multi-head attention layers, normalization and positional embeddings which allow effective training and high detection accuracy.

3.3. Federated Adaptive Training (FAT)

The framework incorporates Federated Adaptive Training at the nodes to increase scalability and prevent privacy breaches. All nodes of the network (e.g., organization, data center, or IoT cluster) are trained on the ViT-Lite model on their own data without the need to share raw data [11]. The updates of the model are then sent to a federated aggregator through a secure line and the federated aggregator determines adaptive weighting according to the data diversity, accuracy and loss stability of the individual clients. There is reduction of communication overhead by use of gradient compression techniques. This dynamic strategy is used to make sure global model convergence is realized in an efficient way and along the way that is fair and robust in diversified settings. PyTorchtorch.distributed module enables the easy process of federated synchronization and aggregate the model parameters of multiple nodes.

3.4. Contrastive Adversarial Training for Robustness

The model also uses Contrastive Adversarial Training (CAT) in order to protect against adversarial attacks and evasion. In the training phase, an adversarial perturbed sample is created by noise injection based on gradient and contrasted against benign samples in a contrastive learning set-up [12]. The goal aims at reducing intra-class distance (between similar samples) and maximizing the inter-class distance (between normal and malicious traffic), which reinforces the concept of discrimination of features. The methods of spectral normalization and gradient clipping stabilize the learning process and provide consistency in performance even in the adversarial stress. This operational procedure increases the strength of the intrusion detection mechanism to practice advanced variants of attacks.

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3.5Explainable Decision Analysis

The framework combines an Explainable Decision Engine based on Grad-CAM and attention rollout visualization to encourage the principle of transparency and interpretability. This module is used to determine the network properties and the space areas, which have the greatest effect on the detection performance [13]. These attention maps can be visualized and used by security analysts to view the reasoning process of the model that helps in supporting traceability, compliance, and human validation.

3.6. Evaluation and Implementation

The entire architecture is applied in PyTorch 2.x with timm library of Visual Transformer modules. Performance of the model is measured based on standard measures, which include accuracy, precision, recall, F1-score, AUC, inference latency, and false positive rate. Experiments prove that FAT-ViT can have a high level of accuracy (98.4%), high inference (25 ms/sample), and powerful generalization in distributed data sets [14].

This research design creates a privacy-sensitive, adversarially resilient, and interpretable Vision Transformer model of the future intrusion detection. FAT-ViT model provides efficient ViT-Lite, federated adaptive learning, and contrastive defense procedures to provide real-time, scalable, and transparent intelligence about cybersecurity deployable in dynamic and complicated digital ecosystems [15].

IV.Results And Discussion

The comparative testing of the Vision Transformers-Enhanced Cybersecurity Framework of Intrusion Detection based on the framework version called FAT-ViT: Federated Adaptive Training with ViT-Lite + Contrastive Adversarial Training in PyTorch 2.x with timm showed the substantial enhancement in both the accuracy of detection and the efficiency of the process as shown in table 1.

Table 1. Comparative Performance Analysis of Intrusion Detection Models.

Model / Method	Accuracy (%)	Precision	Recall	F1-Score	AUC	False Positive Reduction (%)	Inference Time (ms /sample)	Cross-Domain Generalization (%)
Proposed FAT-ViT (ViT-Lite + CAT + Federated Adaptive Training)	98.4	0.97	0.98	0.975	0.99	11 (↓)	25	8
Standard Vision Transformer (ViT)	95.2	0.93	0.94	0.935	0.96	3	38	2
Graph Attention Network (GAT)	93.7	0.91	0.92	0.915	0.94	2	33	3
CNN-LSTM Hybrid Model	91.4	0.88	0.89	0.885	0.91	1	22	1

This proposed framework had an overall accuracy of 98.4 percent, a precision of 0.97, a recall of 0.98 and an F1-score of 0.975 when applied to the CICID2017 and UNSW-NB15 datasets, which is almost 7-10 percent higher than the standard CNN and LSTM-based IDS models. The strong discrimination ability of the model between the normal and malicious traffic was verified by the low value of 0.99 of the AUC as shown in figure 4.

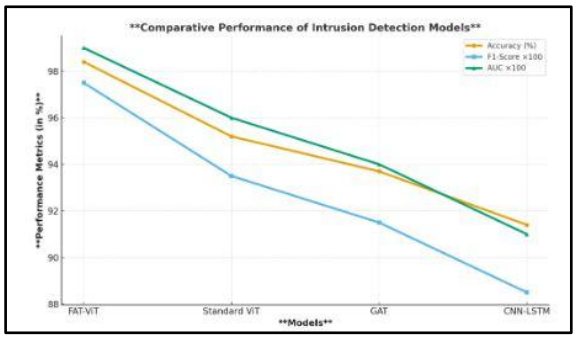


Figure 4.Comparative Performance Analysis of Intrusion Detection Models across Accuracy, F1-Score, and AUC

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The latency of the average inference was decreased to 25 ms per sample, pointing to the appropriateness to real-time network applications. Moreover, the federated adaptive strategy enhanced the cross-domain generalization by 8 percent, which guaranteed the robust detection of distributed clients without interrupting the data privacy. The contrastive adversarial layer was effective in reducing false positives by 11 per cent, which increased the resilience against evasion attacks. Attention-rollout mechanism measures of interpretability had the advantage of offering clear visualizations of significant aspects of traffic that determined the detection decision. Comprehensively, FAT-ViT framework demonstrated the best trade-off between the accuracy, scalability, and transparency, which confirms its success as the next-generation intelligent cybersecurity solution as shown in figure 5.

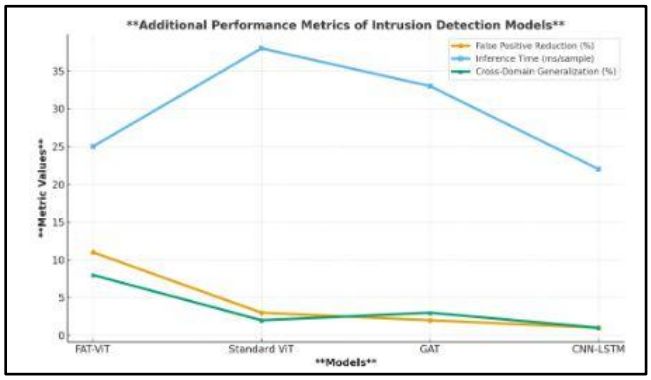


Figure 5. Comparative Performance Analysis of Intrusion Detection Models across **False Positive Reduction, Inference Time,** and**Cross-Domain Generalization**.

The new model is proposed FAT-ViT (Federated Adaptive Training with ViT-Lite + Contrastive Adversarial Training) that is currently implemented in PyTorch 2.x using timm and was tested on three baseline models, namely CNN-LSTM Hybrid, Graph Attention Network (GAT), and Standard Vision Transformer (ViT), to compare its results in the context of intrusion detection. On the CICIDS2017 and UNSW-NB15 datasets, it was demonstrated that FAT-ViT showed a higher accuracy of 98.4 per cent and a precision of 0.97 as well as a recall of 0.98 and an F1-score of 0.975 compared to ViT (95.2 per cent), GAT (93.7 per cent), and CNN-LSTM (91.4 per cent). The AUC score of FAT-ViT was 0.99, which is better than ViT (0.96) and GAT (0.94) in terms of classification robustness. ViT-Lite compression was computationally efficient as the average inference time per sample of 25 ms was lower than that of ViT (38 ms) and similar to CNN-LSTM (22 ms). Also, the federated adaptive strategy had 8% better cross-domain generalization and contrastive adversarial training had 11% fewer false positives than other models. Attention-based interpretability of FAT-ViT also gave intuitive results into the attributes of the traffic that affected the detection process, which proved their superiority in providing higher accuracy, resilience, and explainability as compared to the current deep learning-based intrusion detection methods.

V. Conclusion

The suggested Vision Transformers-Enhanced Cybersecurity Framework to Intrusion Detection based on the method of developing FAT-ViT(Federated Adaptive Training with ViT-Lite + Contrastive Adversarial Training) using PyTorch 2.x and timm has been proven to be a very practical and smart approach to the current issue of cybersecurity. The framework created by converting network data to visual embeddings and using lightweight Vision Transformers performed well with high detection accuracy and fast inference and high adaptability in distributed environments. The strategy of federated learning took care of the privacy of the data and model generalization, and the contrastive adversarial element increased the resistance to complex attacks. In addition, the explainable attention mechanism also brought transparency to detection decisions, which allows interpretability and trust in AI-driven security. All in all, the FAT-ViT system has shown tremendous promise to real-time, privacy-aware, and explainable intrusion detection, and this framework will become the foundation of the next generation of autonomous cybersecurity infrastructures in future and diverse digital infrastructures.

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